

System for Visual Detection of Individual Dosimeters Using Convolutional Neural Networks

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1. INTRODUCTION

Radiological safety is essential in facilities operating with ionizing radiation sources. Individual monitoring with dosimeters plays a key role in quantifying workers' absorbed dose. In Brazil, CNEN Standard NN 3.01 (Resolution 323/24) requires individual monitoring of exposed workers [1], but program effectiveness may be compromised by operational failures, misuse, or absence of devices at the time of exposure.

Advances in artificial intelligence, particularly in computer vision, have enabled real-time identification of specific objects. Convolutional neural networks (CNNs) stand out for visual recognition and are increasingly applied in the nuclear sector. This work presents an automated system for visual detection of individual dosimeters using YOLOv8 to complement traditional personal dosimetry with continuous, automated verification of proper device use.

2. MATERIALS AND METHODS

The system was developed in Python using Google Colab, employing the YOLOv8 architecture. Images were annotated in Roboflow, with the dataset split into 70% training, 20% validation, and 10% testing.

A total of 723 images from the Laboratório de Instrumentação e Detecção Nuclear at IME, Rio de Janeiro, were used, featuring three types of dosimeters: TLD (Thermoluminescent), Film, Pen.

Figure 1 presents an image with the three types of dosimeters used in this study, properly identified by name and class.



Figure 1 – Three types of dosimeters used in the study, identified by class.

Figure 2 shows an example of how the images were labeled during the annotation process.



Figure 2 – Example of image annotation in Roboflow Tool.

Figure 3 illustrates the conceptual model of convolutional neural networks in the classification of the dosimeter, while Table 1 presents the training parameters used.

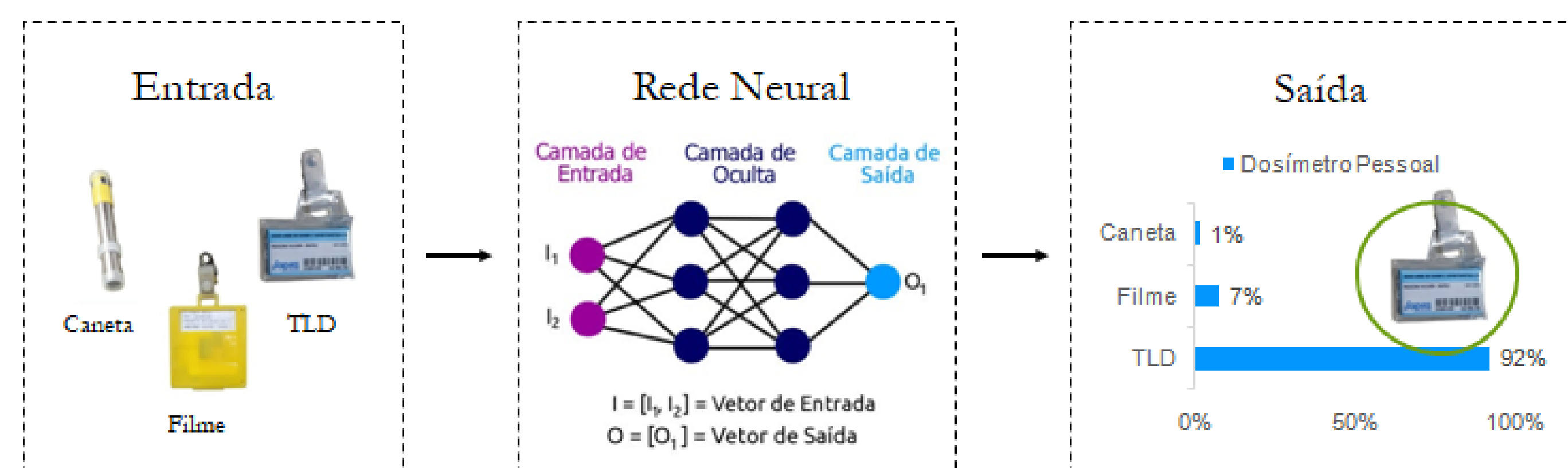


Figure 3 – Simplified schematic of neural network operation

Table 1 – YOLOv8 Model Training Settings

Parameter	Description	YOLOv8
Number of epochs	Number of training iterations	50
Image size	Image dimensions	640 x 640
Batch size	Number of images per batch	16
Number of classes	Detection classes	3 (TLD, Film, Pen)
Architecture	Model type	YOLOv8
Optimization method	Optimization algorithm	AdamW

3. RESULTS

Table 2 presents the main performance indicators of the automated individual dosimeter detection system using YOLOv8:

Table 2 - General Results of the YOLOv8 Model for Dosimeter Detection

Metric	Value
Precision	0.889
Recall	0.969
mAP@0.5	0.977
mAP@0.5:0.95	0.715
Fitness	0.742
Inference time (ms)	0.900
Post-processing time (ms)	2.720

4. CONCLUSIONS

This study demonstrated the feasibility of using convolutional neural networks for automated visual detection of individual dosimeters. YOLOv8 achieved high accuracy, low latency, and potential for real-time use on embedded devices. The proposed solution strengthens existing radiological protection systems by adding an extra layer of control to reduce human error and improve the reliability of occupational exposure records.

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